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Database Management System

Assignment - 1

Ques-1 What is DBMS? Why is it required?

⇒ DBMS (Database Management System) is software that stores, manages & retrieves data in an organized way. It is required to reduce data redundancy, maintain consistency, provide security & allow easy data access.

Ques-2 Why is DBMS better than traditional file system?

⇒ DBMS reduces duplication of data, avoids inconsistency, provides security, supports multiple users at the same time & offers backup & recovery facilities.

Ques-3 What is Data Abstraction? Explain its levels.

⇒ Data abstraction means hiding internal details & showing only necessary information.
Three levels are :-

- Physical Level - How data is stored.
- Logical Level - What data is stored & relationships
- View Level - How users see the data.

Ques-4 What is Three Schema Architecture? Why is it important?

⇒ Three-schema architecture includes Internal, Conceptual & External levels. It is important because it provides data independence & separate user views from physical storage.

Ques-5 What is Data Independence? Why is it needed?

⇒ Data Independence means changes at one level of database do not affect other levels. It is needed for flexibility & easy modification of the system.

Ques-6 What is ER Model?

⇒ ER Model is a conceptual diagram that represents entities, their attributes & relationships b/w them in a graphical form.

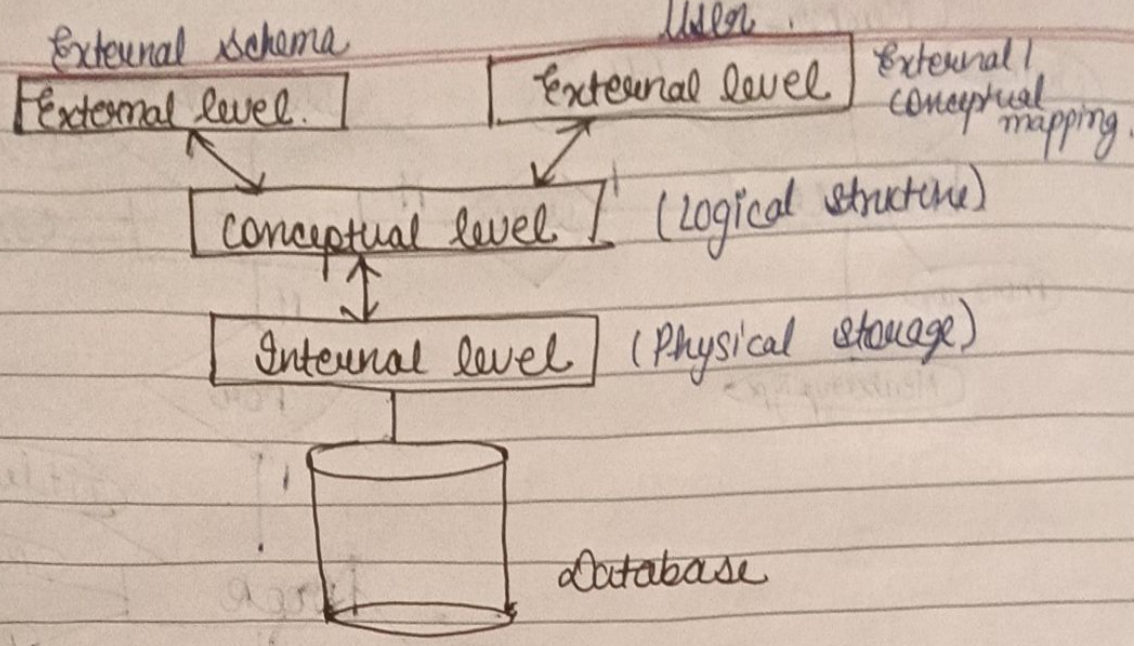
Ques-7 What are Integrity constraints?

⇒ Integrity Constraints are rules that maintain accuracy & consistency of data.

Example :- Primary key, Foreign key, Unique & Not Null constraints.

Ques-# Section-B

Ques-8 Draw the three-Schema architecture & explain each level.



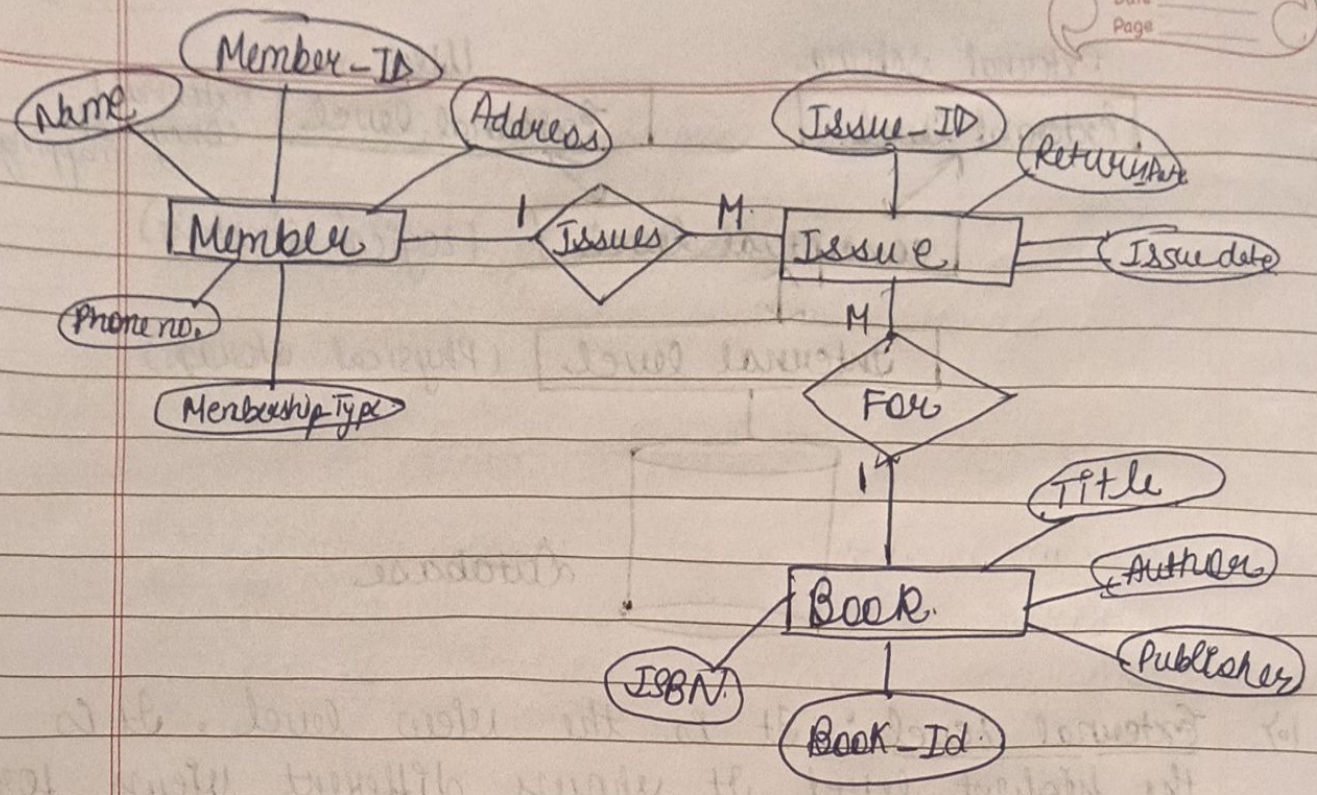
1.) External Level :- It is the view level. It is the highest level. It shows different views for different users. It hides complete database completely.
ex:- Student see marks only.

2.) Conceptual level : Describes complete database structure. It Includes :
- Entities
- Attributes
- Relationships
- Constraints.

⇒ Independent of Physical storage.

3.) Internal level :- (Physical level) :- Lowest Level. It describes - how data is stored, Indexing, file structure, storage blocks.

Ques. Draw an ER diagram for a Library Management System.



Relationships :-

⇒ Member – Issues – BOOK.
⇒ Relationship is Many-to-Many
⇒ One member can issue many books.

Cardinality :-

1 : M
M : 1

Ques Draw an ER diagram for a student-course Registration System.

⇒ Student (Student ID, Name)
Course (Course-ID, Course Name)

Relationship :-

Student register for course. (M-M)

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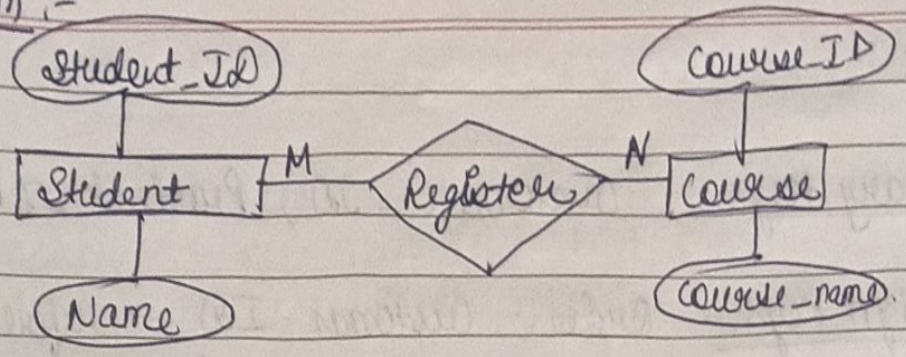
ER Diagram :-



Cardinality :- (M:N).

Section - C

Ques A hospital wants to store patient, doctor & appointment data. Identify entities, attributes & relationships.

⇒ Entities :- Patient, Doctor, Appointment.

Patients Books appointment ⇒ Relationship

Doctor attends appointment ⇒ Relationship

Primary Keys :-

Patient - ID, Doctor - ID, Appointment - ID.

Ques An online shopping platform stores customer, product & order data, suggest Keys & constraints.

Entities :- Customer :-

① Customer_ID, Name, Email (Unique), Phone, Address.

② Product :- Product_ID, Product_Name, Price, Book Quantity, category.

③ Order :- Order_ID (Primary Key), Order_Date, Total Amount, Customer_ID (F.K.)

Keys :-

① Primary Keys :- Customer_ID, Product_ID, Order_ID

② Foreign Keys :- Order :- Customer_ID → references Customer (Customer_ID)

Constraints :-

(i) Entity Integrity Constraints :- Primary key can't be NULL
ex :- Customer_ID ~~must~~ must have a value

(ii) Referential Integrity Constraints :- Foreign key must match an existing primary key.
ex :- You can't place an order for a non-existing customer.

(iii) Domain Constraints :- Data must follow correct data type & range.
ex :- Price > 0

(iv) Unique Constraint :- Email should be unique.

(v) Check constraint :- Quantity > 0
Total_amount ≥ 0

Ques A university stores student records. Explain how data abstraction helps different users.

⇒ Data abstraction means hiding unnecessary details & showing only required information.

to users.

1) View Level (External Level) :- Diff. users diff. views

- Student - sees marks & attendance.
- Teacher - sees student list & grades.
- admin - sees full records.
- Accounts - sees fee details only.
- Provide security.

2) Logical level :- Entities :- (Student, Course, Faculty)
Attributes :- (Student_ID, Name, DOB)
Relationships :- (Enrolls, Teaches)

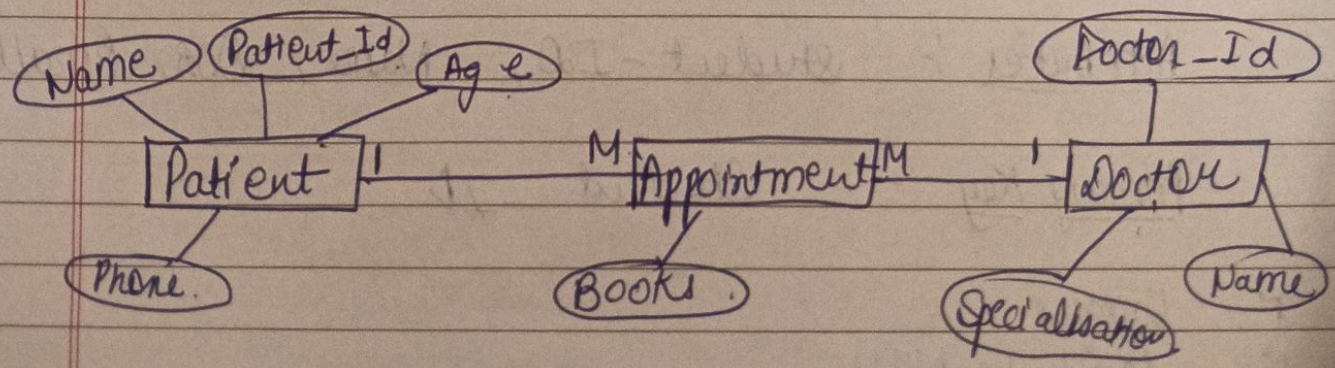
3) Physical level :- Describes how data is stored.
- Indexing
- File organization
- Storage blocks.

Section-D :-

Ques Draw a simple ER diagram for any real-life system & explain one key design decision.

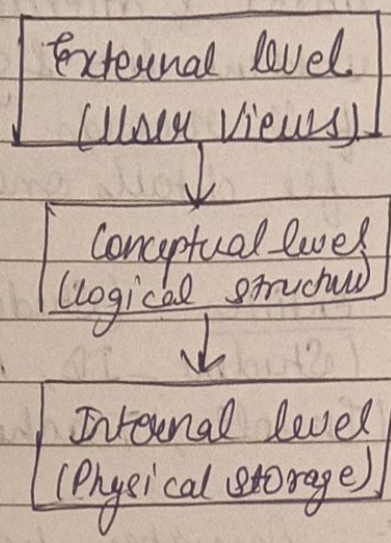
Hospital Management System :-

Entities :- Patient, Doctor, Appointment.



Ques Draw the three - schema architecture & mention one benefit.

⇒



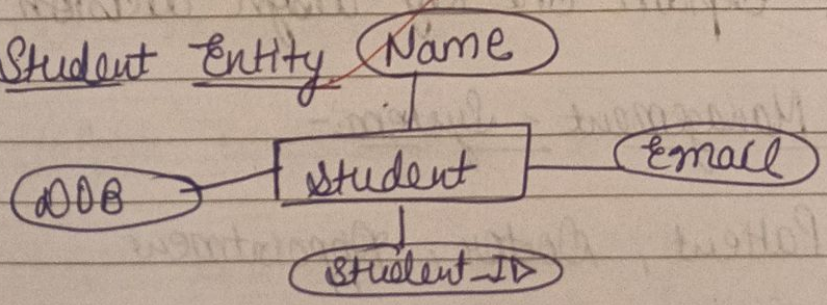
One Benefit :- **Data Independence** :-

Changes at Physical level → do not affect logical level.
logical level → do not affect user views

↳ This makes system flexible & secure.

Ques Draw an entity with attributes & identify its primary key.

⇒ **Student Entity**



Attributes :- Student-ID, Name, DOB, Email.

Primary Key :- Student_ID

↳ Unique
↳ Can't be NULL.
↳ Identifies each student uniquely.